1 Objectives and Significance

The heightening of concerns about national security has intensified the need for effective, efficient surveillance. The ability to find images and video sequences that fit a query profile in a massive surveillance database offers a strategic advantage to those who can assemble such information quickly and accurately. This project aims to advance key technologies, *feature extraction* and *event mining*, that enable such searches to be both accurate and efficient.

The first research thrust, *feature extraction*, deals with finding spatial and temporal cues from images and videos for depicting target events. In addition to extracting representative perceptual features, this thrust must deal with the *inconsistent feature* problem caused primarily by two sources.

- *Different camera capabilities*. Different cameras generate videos of different color intensity and texture quality. Furthermore, camera movements (e.g., panning and zooming) can affect video features, as can the distance between cameras and the areas of surveillance.
- *Variable environment factors*. External environment factors such as lighting conditions (e.g., day or night) and weather (e.g., foggy or raining) also affect video features.

The second research thrust, *event mining*, deals with mapping low-level features to high-level events. Event mining faces two major challenges for conducting statistical inference.

- *Imbalance in training data*. Positive events (hazardous events) are always outnumbered by negative events in the training data. In an imbalanced set of training data, the class boundary tends to skew towards the minority class and becomes very sensitive to noise. Hence the rate of false negatives increases.
- *Scarcity of training data*. Let D denote the number of data dimensions, N the number of training instances. Event mining faces the $D > N$ high dimensionality challenge. As pointed by D. Donoho [?], the theories underlying "classical" data analysis are based on the assumption of $D < N$, and $N \rightarrow \infty$, but when $D > N$, the basic methodology which was used in the classical world is not strictly applicable.

Intellectual Merit

To address the challenges in feature extraction and event mining, and to ensure that our proposed solutions are scalable, this project's specific targets are as follows:

1. To devise *invariant descriptors* for improving visual data characterization. (See tasks A, B, and C in Section ??.)

2. To design *learning algorithms* that can thoroughly profile a target event despite limited training data and the dimensionality curse. In addition to developing learning algorithms that works under the $D > N$ constraint, we will also develop novel dimension reduction techniques. (See tasks D, E, and F in Section ??.)

3. To conduct *validation* on developed learning algorithms with experimental data provided by colleagues at IBM Almaden Research Center, IBM T.J. Watson Research Center, and SONY. (See Section ??.)

PIs' Prior Accomplishments

The learning algorithms (MEGA [?, ?] and SVM_{Active} [?]) and perceptual distance function (DPF [?, ?]) that Edward have recently developed provide a good foundation for our research team to continue making progress in high-dimensional scarce-data statistical inference. In addition, the prototype [?, ?] that we have implemented allows us to quickly examine new hypotheses (some are described in Section ??), and to solicit input from the research community and from our partners.

Broader Impacts

The broader impacts of this work will be substantial because that our two research thrusts tackle fundamental challenges in computer vision and machine learning, respectively. In addition, the problem at hand is the national priority. Any major advancement in national security will contribute to peace and save lives. Specifically, we summarize our project impacts as follows:

1. Scientific advancement. The learning algorithms that we will develop for working under the $D > N$ constraint can benefit many emerging applications. For instance, in the application of gene profiling, number of genes is much larger than the number of clinical samples can be obtained to conduct statistical analysis.

2. National security. Being able to identify target events accurately and quickly will certainly be helpful to intelligence agencies, who have been overwhelmed by the huge amount of available data.